

Gradient Boosting

A powerful ensemble ML technique that builds a strong predictive model in a sequential, stage-wise fashion by combining multiple weak learners, typically decision trees.

It minimizes errors by training new models to predict the residuals (errors) of previous models, often using gradient descent.

Key concepts and Working Mechanism

1) Sequential Learning

Unlike random forest, which builds trees in parallel, Gradient Boosting builds trees sequentially, where each new tree corrects the errors (residuals) of the

combined ensemble of previous trees.

2) Loss Function Optimization

The algorithm attempts to minimize a defined loss function (eg MSE for regression) by adding weak learners, effectively moving down the gradient.

Shrinkage (Learning Rate)

To prevent overfitting, the contribution of each weak learner is scaled by learning rate ($\lambda < 1$), requiring more trees but often resulting in better generalization.

Weak Learners

Decision trees are the standard weak

learners, but other models can be used.

			m_1		m_2		m_3	
IQ	Cgpa	Salary	Pred1	r_1	Pred2	Res2	Pred3	
90	8	3	4.8	-1.8	-1.8	-1.62	-1.62	
100	7	4	4.8	-0.8	-0.8	-0.72	-0.72	
110	6	8	4.8	3.2	3.2	2.88	2.88	
120	9	6	4.8	1.2	1.2	1.08	1.08	
80	5	3	4.8	-1.8	-1.8	-1.62	-1.62	

① M_1

Step1 - Calculate Prediction (Pred1)

Pred1 is calculated by taking mean of y

Step2 - Calculate Error

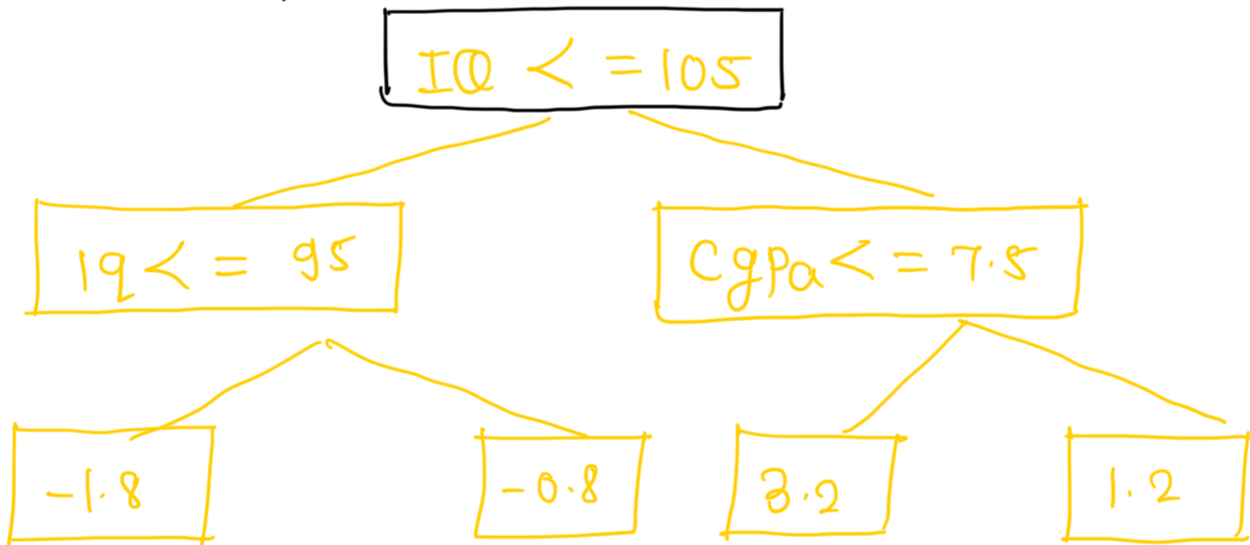
Residual (r_1) = Actual - Pred = Pseudo residual.

② M_2

Model 1 only calculates the mean

Model 2 - Learns from errors of m_1 & forms a decision tree

Takes input columns & res_1

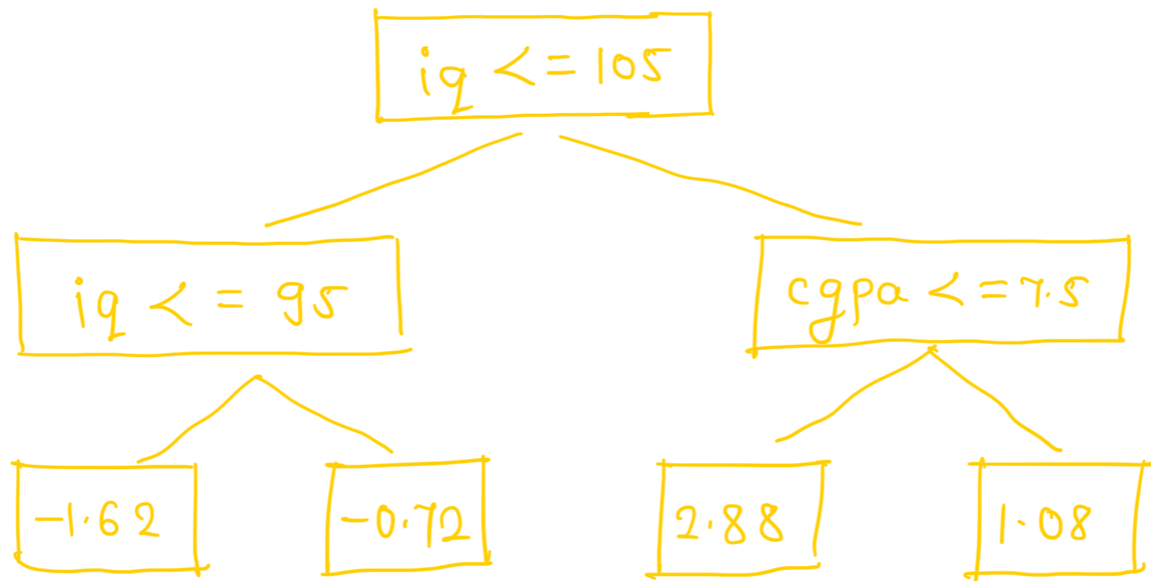


$$Y_{\text{pred}_2} = m_1 + 0.1 \times m_2$$

$$\begin{aligned} \text{Res}_2 &= \text{Actual} - \text{Predicted} \\ &= 3 - (4.8 + 0.1 \times (-1.8)) \\ &= -1.62 \end{aligned}$$

③ M_3

Takes input columns and res_2 & creates a decision tree



$$Y\text{-pred} = m_1 + 0.1 \times m_2 + 0.1 \times m_3$$

Input $\Rightarrow iq = 60, cgpa = 4.9$

Prediction $\Rightarrow m_1 + 0.1 \times m_2 + 0.1 \times m_3$

$$= 4.8 + 0.1 \times (-1.8) + 0.1 \times (-1.62)$$

$$= 4.45$$